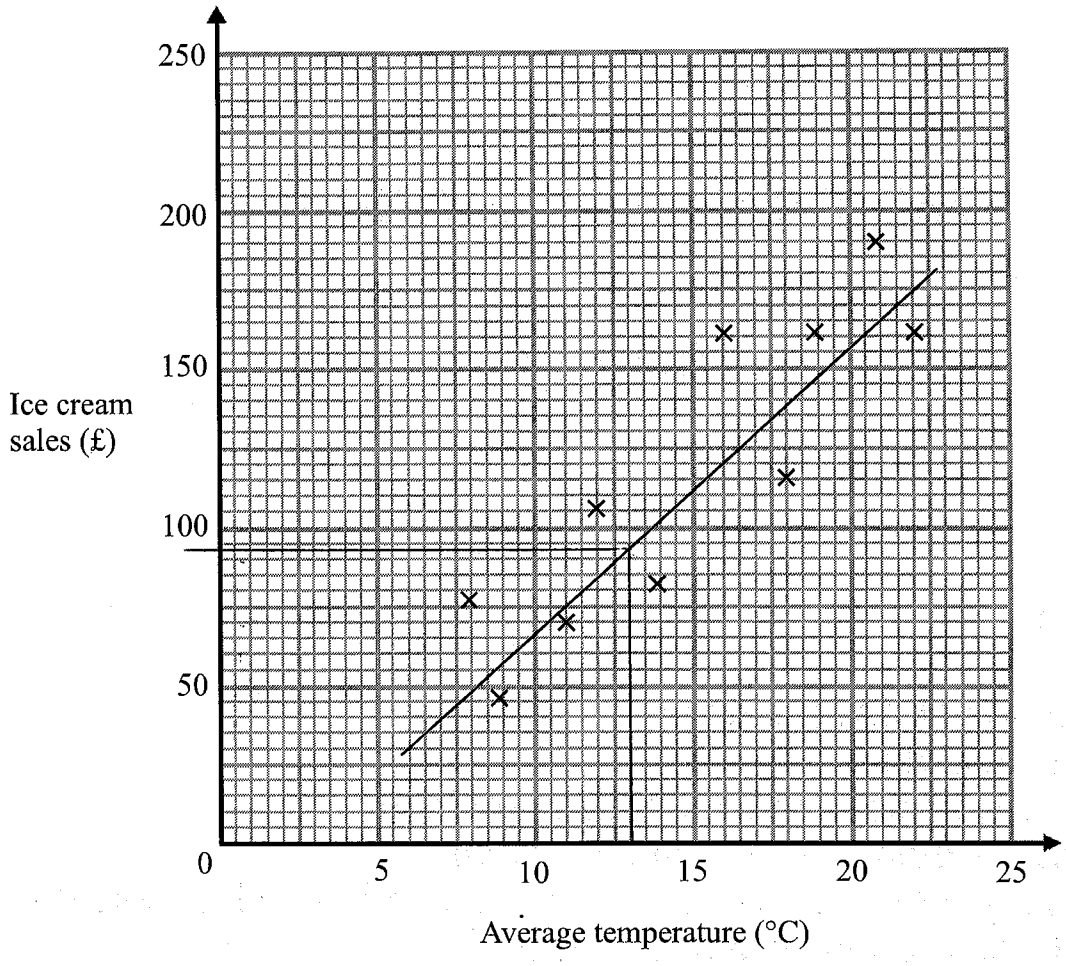
1 The average daytime temperature for 10 days is recorded.
A shop also records its ice cream sales for each of the 10 days.

The scatter graph shows this information.



(a) What type of correlation does the scatter graph show?

positive

(1)

(b) On the 11th day the temperature was 13°C.
Estimate the ice cream sales on the 11th day.

£90

[90 – 110] (2)

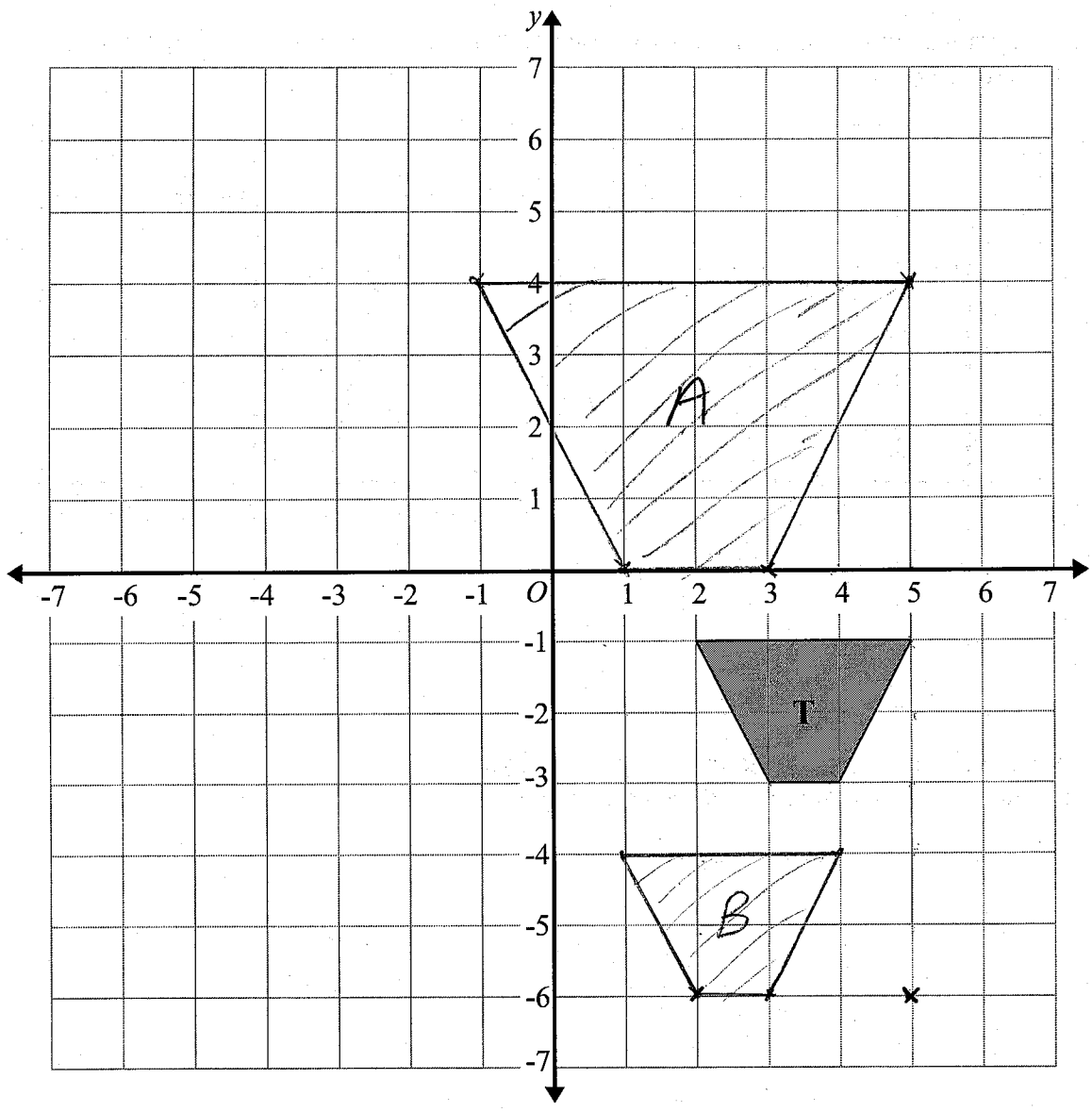
(c) The shop's manager wants to use the scatter graph to predict the ice cream sales for a day with an average temperature of 2°C. Comment on the reliability of this prediction.

It would be unreliable – 2°C is out of the range of data

(1)

(Total for Question 1 is 4 marks)

2

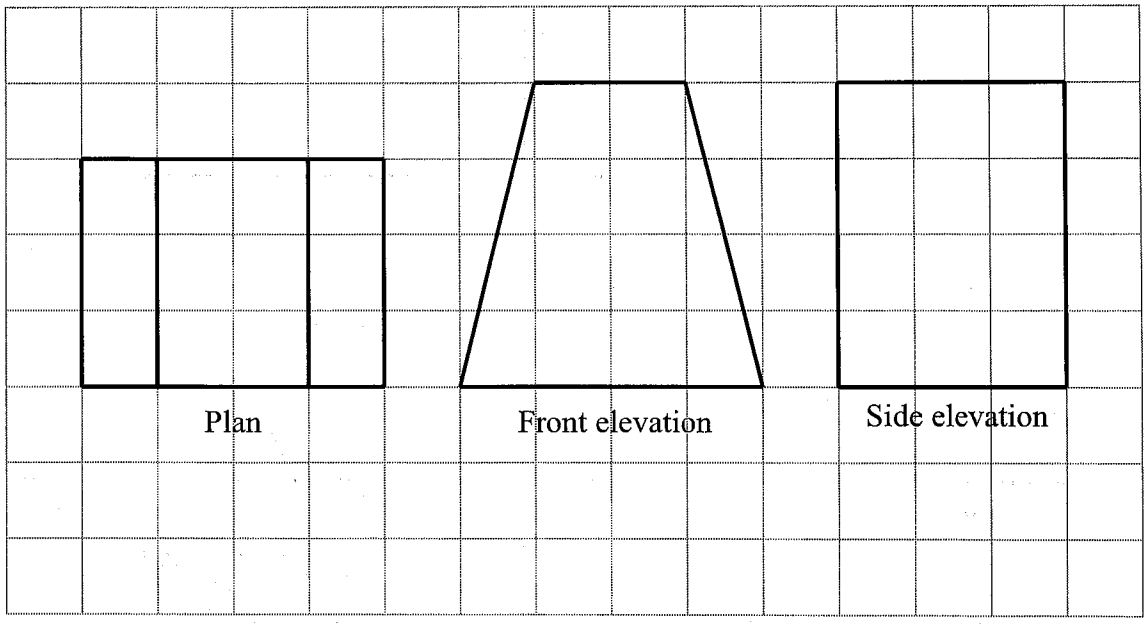


(a) Enlarge trapezium **T** by scale factor 2 centre (5, -6).
Label the new trapezium **A**.

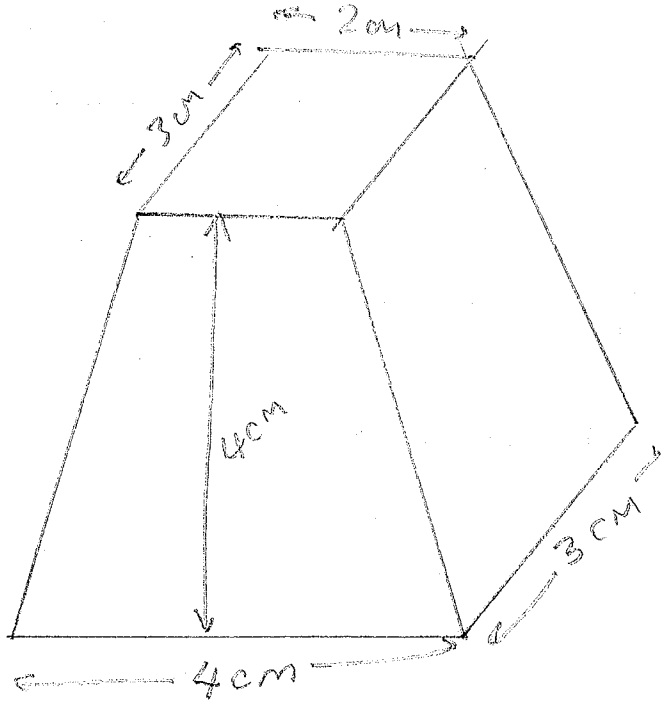
(b) Translate trapezium **T** by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$
Label the new trapezium **B**.

(Total for Question 2 is 4 marks)

3 The diagram shows the plan, front elevation and side elevation of a solid shape, drawn on a centimetre grid.



In the space below, draw a sketch of the solid shape.
Give the dimensions of the solid on your sketch.



(Total for Question 3 is 2 marks)

4 (a) Expand $2x^2(x + 5)$

$2x^3 + 10x^2$

(2)

(b) Expand and Simplify $4(y + 7) - 2(y - 6)$

$4y + 28 - 2y + 12$

$2y + 40$

(2)

(Total for Question 4 is 4 marks)

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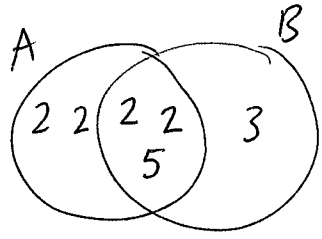
5 $\mathbf{A} = 2^4 \times 5$ $\mathbf{B} = 2^2 \times 3 \times 5$

(a) Write down the highest common factor (HCF) of **A** and **B**.



$2 \times 2 \times 5$

4×5

20

(1)

(b) Find the lowest common multiple (LCM) of **A** and **B**.

$2 \times 2 \times 20 \times 3$

$4 \times 20 \times 3$

80×3

240

240

(2)

(Total for Question 5 is 3 marks)

6 (a) Given $\dfrac{x^8}{x^a} = x^{11}$

Find the value of a.

$8 - a = 11$

$8 = 11 + a$

$-3 = a$

$a =$ -3

(1)

(b) Simplify $(5m^4)^3$

$125m^{12}$

(2)

(Total for Question 6 is 3 marks)

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7 Work out an estimate for the value of $\dfrac{3.14 + \sqrt{98.9}}{0.19}$

$\dfrac{3 + \sqrt{100}}{0.2}$

$\dfrac{3 + 10}{0.2}$ $\quad \dfrac{13}{0.2}$ $\quad 13 \times 5 = 65$

65

(Total for Question 7 is 3 marks)

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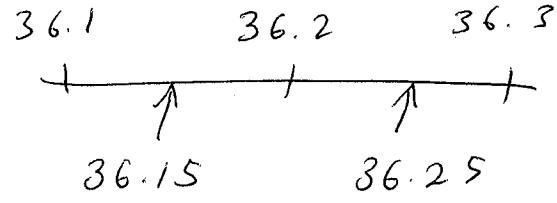
8 A number x is rounded to 3 significant figures.

The result is 36.2

Write down the error interval for x.



$36.15 \leqslant x < 36.25$

(Total for Question 8 is 2 marks)